AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

AKU EAST AFRICA EXAMINATIONS 2016

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

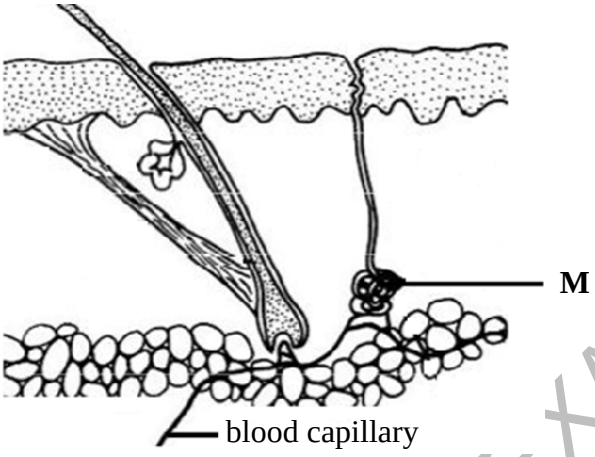
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. Write TWO similarities between photosynthesis and respiration. (2 Marks)

b. The diagram shows a section of the human skin.



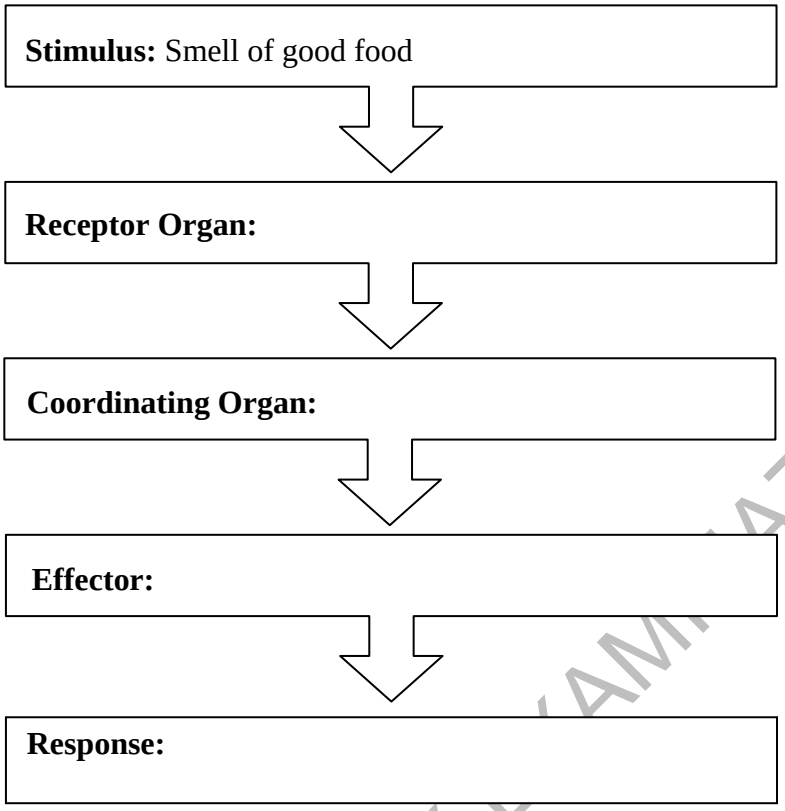
Briefly describe the role of the labelled structure **M** in keeping the body temperature constant.

(2 Marks)

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Q.2. (Total 4 Marks)

Complete the given flow chart with reference to the structures involved in the given example of nervous coordination.



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Q.3. (Total 4 Marks)

a. Briefly describe the working of ball and socket joints. (2 Marks)

_____

_____

_____

_____

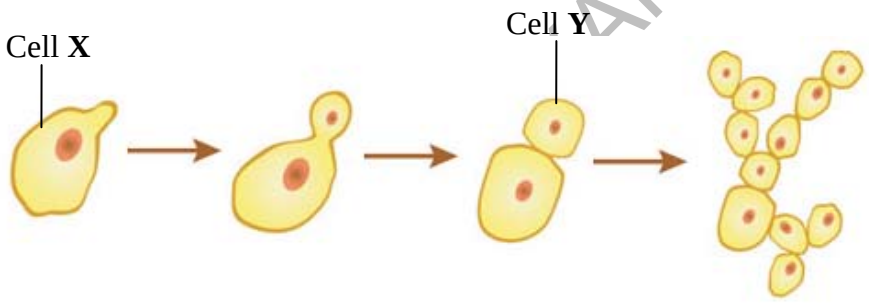
b. Name the type of connective tissue present in the human body to join: (2 Marks)

Muscles to bones _____

Bones to bones _____

Q.4. (Total 4 Marks)

a. The diagram below shows a type of reproduction in an organism.



Is cell **Y** genetically different from cell **X**? Give ONE reason to support your answer. (2 Marks)

_____

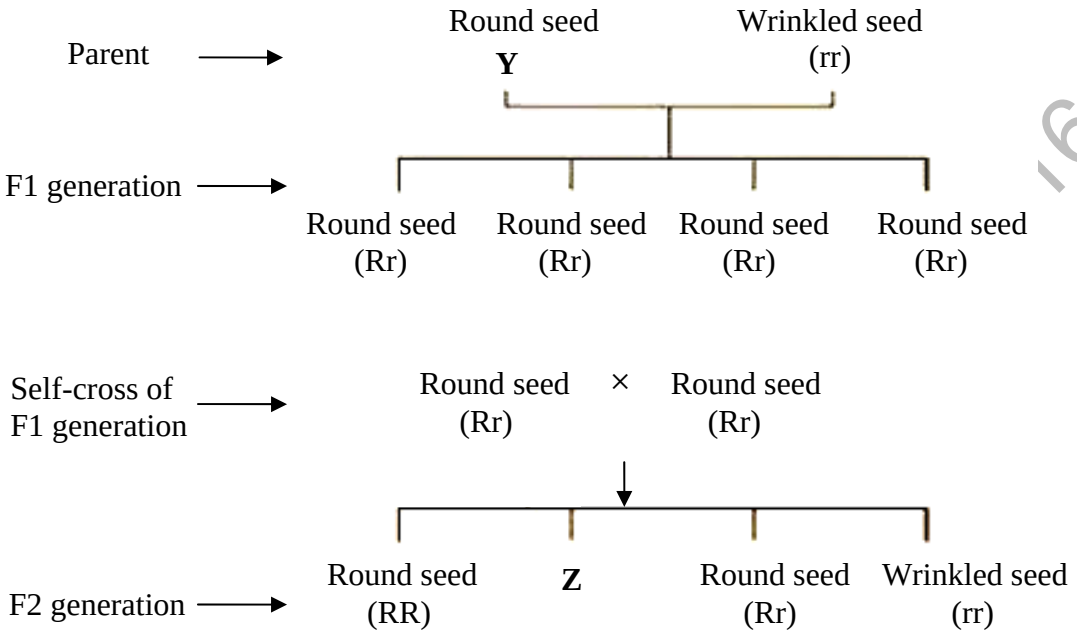
_____

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b. The flow chart below shows the results of the crosses between different seeds of a plant species.



Find out the missing genotypes **Y** and **Z** in the given case. (2 Marks)

Genotype **Y**:_____

Genotype **Z**:_____

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Q.5. (Total 4 Marks)

a. Why is the flow of energy considered as non-cyclic in an ecosystem? (2 Marks)

_____

_____

_____

_____

b. Briefly describe the process through which nitrogen is returned to the atmosphere.
(Hint: Refer to the nitrogen cycle) (2 Marks)

_____

_____

_____

_____

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Q.6. (Total 5 Marks)

a. Briefly describe the process of fermentation by bacteria. (2 Marks)

_____

_____

_____

_____

b. Given below is a set of drugs.

Mescaline	Aspirin	Diazepam	Psilocin

Identify hallucinogens and sedatives from them. (3 Marks)

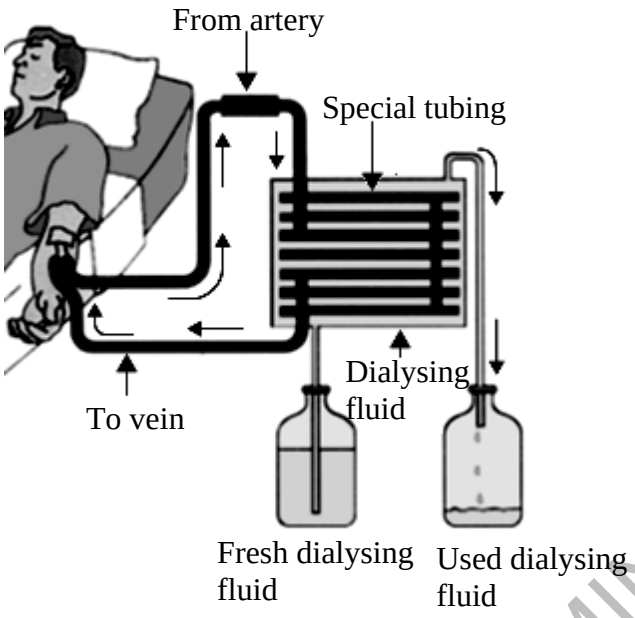
Hallucinogen	Sedative

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Q.7. (Total 7 Marks)

EITHER

a.



i. Name the type of dialysis shown in the diagram.

ii. Dialysis machine is considered to be an artificial kidney. Describe the features of a dialysis machine on the basis of which it can substitute the functions of different parts of the nephron.

OR

b. Select ONE example each of nervous and chemical coordination from your daily life and describe them to highlight the significance of coordination in the body.

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Q.8. (Total 8 Marks)

EITHER

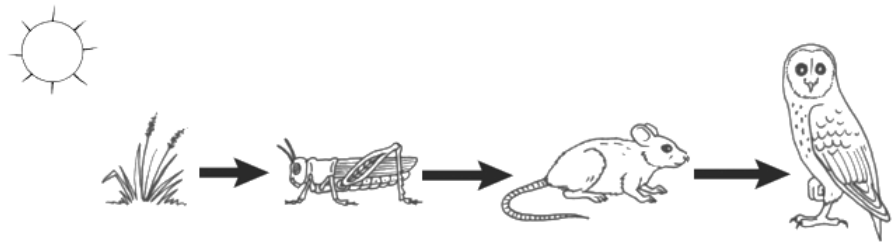
a.

i. How are hypogeal and epigeal germination different from each other?

ii. State THREE ideal conditions which can help a seed to germinate successfully. Also describe the importance of each condition.

OR

b.



i. Compare the flow of energy from grass to owl in the above food chain assuming that there is 10,000 KJ energy at the producer level.

ii. Construct a pyramid of biomass of the given food chain.

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Space for diagram

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END OF PAPER

Please use this page for rough work

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